

WHY ARE INSECTS DECLINING _ XIAOQI WU.pdf

by Yuki Wu

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Coursework Coversheet

Name	Xiaoqi (Yuki) Wu
Date	21/06/2024
Module	Tutored Dissertation
Coursework item	Tutored Dissertation

Self-reflection is a non-assessed part of the coursework submission to aid your learning

1. Have you used previous feedback/self-reflection to produce this coursework? If so, how? If not, why?

Yes, I utilised my previous feedback to improve my time management by creating a well-organized schedule for completing this essay.

2. What do you feel went well? (e.g., development of specific skills, sense of achievement, part of the assignment you are most proud of)

I have improved in paraphrasing and finding relative materials.

3. What do you feel did not go well? (e.g., learning outcomes that you found most difficult to meet, aspects of the assignment that were challenging, time-management)

What I found challenging was shortening my essay within the word limit.

I did not leave much time to revise my essay, which I think I could improve next time.

Additionally, I am still not that confident with my structure. I am afraid of writing off the topic by not heavily focusing on it, which is explaining why insects are declining.

4. What did you hope to learn from doing this coursework? Did you achieve that?

This is the first time in my life that I have written an essay this long.

I anticipated learning critical thinking skills, which I was bad at before.

I have tried my best to read papers critically and have included many critiques of the topic and papers.

Why are Insects Declining?

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INTRODUCTION

The human population has surged to 8.1 billion (Worldometers.info, 2024), and the Anthropocene hugely impacts Earth's environment and ecosystems. Human activities, such as intensive agriculture, urbanisation, and the spread of invasive species, have led to habitat loss and fragmentation, increased atmospheric CO₂ levels, pollution, climate change, and altered extreme weather and precipitation patterns (Ceballos et al., 2015; Goulson, 2019). These anthropogenic pressures are causing severe biodiversity loss, with predictions that we are entering the sixth mass extinction event (Dirzo et al., 2014).

Anecdotal evidence of insect declines includes fewer insects on car windshields, diminishing firefly displays, and fewer moths observed at gas stations and under streetlights. Scientifically, numerous studies report species loss and population decline, indicating a troubling trend. Insects play crucial roles in ecosystems, including pollination (Ollerton, Winfree & Tarrant, 2011; Vanbergen & Initiative, 2013), nutrient cycling (Losey & Vaughan, 2006; Yang & Gratton, 2014), and serving as a food source for various animals. Their decline could disrupt these essential services, leading to far-reaching implications for biodiversity and human welfare.

Understanding the factors driving this decline is crucial. Various threats have been identified, including habitat loss, pesticide use, climate change, and agricultural practice. These factors often interact synergistically, exacerbating their overall impact on insect populations. Moreover, assessing insect declines presents significant challenges due to reporting biases and the inherent difficulties in studying these diverse and often elusive organisms (Sánchez-Bayo & Wyckhuys, 2019).

This dissertation aims to review and criticise the papers reporting insect decline, explore the primary drivers of insect decline, examine the ecological, economic, and health impacts, and review current conservation efforts. Furthermore, it will critically evaluate the current research and policy actions, identifying gaps and recommending future directions. Recognising that insect decline is not a uniform phenomenon, this dissertation will also examine species thriving amidst changing conditions to uncover potential conservation strategies that can be applied to more vulnerable species.



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HISTORICAL CONTEXT OF INSECT POPULATIONS

Insect populations have experienced significant changes over the centuries, influenced by both natural events and human activities. Before the Industrial Revolution, insect populations were primarily shaped by natural factors such as climate fluctuations, predation, and disease. However, with the onset of the industrial era, human activities became the dominant force driving changes in insect populations (**Fig. 1a**). The widespread adoption of synthetic pesticides, such as DDT, in the early 20th century boosted agricultural productivity but had detrimental effects on **non-target insect**

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species. (Davis, 2018). Historical records show marked declines in beneficial insect populations, such as pollinators and natural pest controllers (Potts et al., 2010; Hallmann et al., 2017). Urbanisation and habitat fragmentation compounded these effects by isolating insect populations, reducing their genetic diversity, and making them more vulnerable to environmental change (Sánchez-Bayo & Wyckhuys, 2019).

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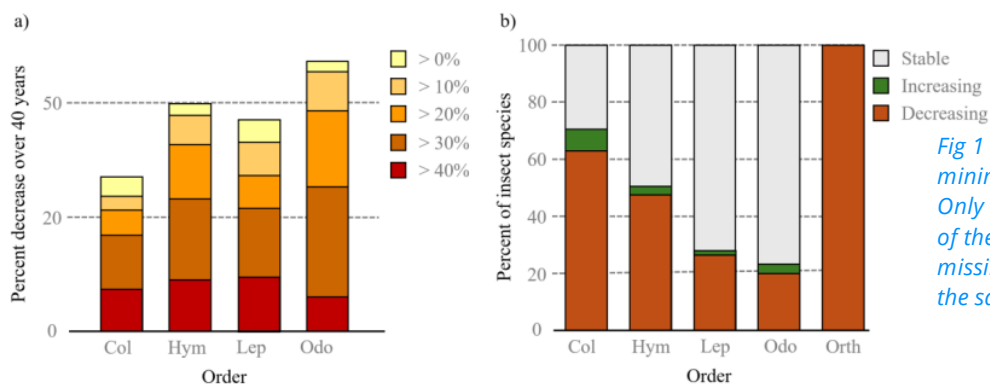


Fig 1 is copied and minimally changed. Only the diagrams of the spp are missing; colours are the same too

Fig. 1 Trends in Insect Species Populations and Declines Over 40 Years, adapted from Dirzo et al. (2014). a) This figure illustrates the percentage decrease in insect species over 40 years across four orders. The data is categorised into different levels of decline, with a significant number of species experiencing over 40% decrease in Hymenoptera and Lepidoptera. b) This figure shows the percentage of insect species that are stable, increasing, or decreasing across five different orders: Coleoptera (Col), Hymenoptera (Hym), Lepidoptera (Lep), Odonata (Odo), and Orthoptera (Orth). The majority of species in most orders are decreasing, with Coleoptera and Orthoptera showing the highest percentages of declining species.

Building on this historical context, recent research has provided a clearer picture of the magnitude and drivers of insect declines. For instance, a comprehensive meta-analysis by Sánchez-Bayo and Wyckhuys (2019) synthesised data from 73 historical reports and concluded that insect **biomass is decreasing at an alarming rate of 2.5% per year**. They also found a staggering 98% decline in ground-dwelling insects over 35 years in Puerto Rico, and over 40% of insect species are declining globally, with a third of species endangered. This study identified habitat loss, mainly due to agricultural expansion, as the primary driver of these declines, **followed by pollution and climate**

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change. Similarly, a significant study conducted in Germany by Hallmann et al. (2017) reported a

75% reduction in flying insect biomass over 27 years in protected nature reserves. This finding is particularly concerning as it highlights severe declines even in areas that are not directly impacted by agricultural activities. Additionally, van Swaay et al. (2022) documented a 35% decline in butterfly populations across Europe from 1990 to 2020, underscoring the widespread nature of these declines. Another study on the decline of carabid beetles in the UK showed significant decreases in these important predator species, which play a crucial role in agricultural pest control (Brooks et al., 2012). Similarly, research on the Dutch moth fauna reported a severe population reduction, aligning with broader trends observed in Europe (Groenendijk & Ellis, 2011).

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While many insect species are experiencing drastic declines, it is important to note that not all insects are declining. Some species are thriving or adapting to changing conditions, highlighting the complex and varied responses of insect populations to environmental changes (**Fig. 1b**). This variation underscores the need for nuanced conservation strategies that consider the specific ecological needs and behaviours of different insect species. The following section will explore these exceptions and discuss the factors contributing to their resilience.

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NOT ALL INSECTS ARE DECLINING

5 Eco-evolutionary dynamics create both winners and losers among insect species. Although meta-analyses on Lepidoptera, one of the best-studied invertebrate taxa, generally indicate a decline in diversity under anthropogenic disturbance, some species remain stable or even increase in diversity in disturbed areas (**Fig. 1b**; Dirzo et al., 2014). For instance, 15 out of 60 species of butterflies in the United Kingdom have expanded their range (Wagner, 2020). Climate change has mixed effects on insect populations; it benefits some species while harming others. For instance, certain species of Britain's moths, particularly those in temperate regions, benefit from alleviated winter limitations by warmer global temperatures. (Hofmann, Fleischmann & Renner, 2018; Hunter et al., 2014). Of the 673 moth species investigated by (Fox et al., 2014), 160 are increasing in abundance and geographic range. Additionally, grass-feeding insects have benefited from the transition of woodlands to savannas or grasslands due to climate change, resulting in increased insect abundance and richness (**7** Sánchez-Bayo & Wyckhuys, 2019). However, species benefiting from anthropogenic impacts tend to be ecologically generalist, whereas stenotopic and specialised species are often adversely affected (Habel, Samways & Schmitt, 2019; Newbold et al., 2018).

Changes in insect abundance vary across ecosystems. A meta-analysis by van Klink et al. (2020) shows that while terrestrial insect populations are declining, freshwater insect populations are increasing. This pattern is supported by studies highlighting the complexity of insect population dynamics across habitats. For instance, grasslands and forests have experienced significant declines in arthropod abundance due to habitat loss and pesticide use (Biesmeijer et al., 2006; Hallmann et al., 2017). Therefore, the variability in responses among different ecosystems underscores the need for

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targeted conservation efforts considering specific ecological contexts (Gaston, Blackburn & Goldewijk, 2003).

Invasive insect species are a prominent example of those benefiting from anthropogenic changes, increasing their abundance and spreading globally through globalisation (Bertelsmeier, 2021). For instance, the Formosan subterranean termite (*Coptotermes formosanus*) invaded Japan 300 years ago through early trade routes; the clonal raider ant (*Ooceraea biroï*) colonised Bangladesh via Bay of Bengal harbours (Trible, McKenzie & Kronauer, 2020); and the tropical fire ant (*Solenopsis geminate*) proliferated worldwide in the 16th century due to Spanish trade routes (Gotzek et al., 2015).

Invasive species often exploit human-induced changes like land development, road construction, and altered hydrologic regimes more effectively than native species (Ros et al., 2020). For example, the gypsy moth (*Lymantria dispar*) and invasive ants benefit from disturbances, establishing dominance in altered habitats (Mack et al., 2000). These invasive species typically possess traits such as rapid colonisation and high reproductive rates, allowing them to outcompete native species in disturbed environments (Catford, Jansson & Nilsson, 2009). However, the success of invasive species can vary depending on the ecological context and type of disturbance. Some invasive species may face challenges like disrupted synchrony with host plants (Lozon & MacIsaac, 1997). This complexity highlights the need for nuanced approaches in managing invasive species tailored to specific environmental conditions.

However, understanding insect population trends is fraught with complexities. Geographical and taxonomic biases and inherent difficulties in accurately assessing these populations obscure the full picture. Despite some species benefiting from anthropogenic changes, our ability to monitor and interpret insect data is limited, making it challenging to draw definitive conclusions about their overall status.

REPORTING BIASES AND DIFFICULTIES OF ASSESSING INSECT POPULATIONS

Geographical and Taxonomic bias

Although these findings revealed notable changes in insect abundance, geographic and taxonomic biases are often ignored. Most research focuses on Western and Northern Europe and North America, with fewer studies from Asia, North America, the Arctic, and the Neotropics (Fig. 2; Wagner, 2020). Insects in wildlands are decreasing but at a slower rate than in Western and Northern Europe, indicating that most reports are biased towards regions experiencing the greatest insect losses. Reports from outside Europe often rely on anecdotal evidence, localised observations, limited taxonomic scope, or impacts from extreme weather events (Wagner, 2020). These biases limit our understanding of insect population trends in underrepresented regions such as Africa, South America, and parts of Asia (Habel, Samways & Schmitt, 2019). Addressing this gap is crucial for developing a global perspective on insect declines.

Studies like the meta-analysis of bees' responses to anthropogenic disturbance (Winfree et al., 2009) are restricted to well-studied taxa and lack sufficient replication and spatial coverage. Research focusing on single taxa, such as using butterflies to indicate the status of European insects (van Swaay et al., 2022), cannot comprehensively represent the local entomofauna (Thomas et al., 2004). Many studies measure total insect biomass to indicate insect decline. However, this method, combined with the oversimplification of complex and stochastic insect population data into higher taxonomic levels, overlooks species-level changes. Since insect biomass is not interchangeable across food webs, this approach fails to analyse ecosystem functions and obscures species-specific trends, leading to a lack of resolution in understanding population declines.

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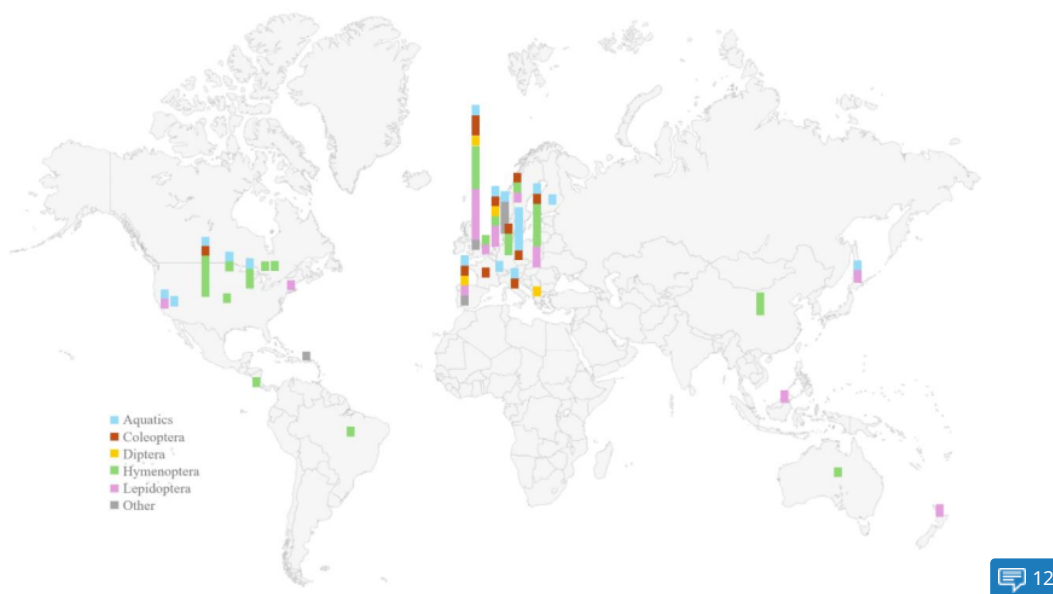


Fig. 2. Geographic locations of the 73 reports, adapted from Sánchez-Bayo & Wyckhuys (2019). The columns represent the relative proportions of surveys for each taxon, as indicated by the different colours in the legend. The data for China and Queensland (Australia) pertain exclusively to managed honeybee populations.

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Difficulties of Assessing Insect Populations

Advancements in research methodologies, including the use of remote sensing and DNA metabarcoding, are enhancing our ability to monitor insect populations more effectively (Kristensen et al., 2018). These technologies enable large-scale, long-term studies that track changes in insect diversity and abundance across various habitats. Despite these advancements, challenges remain in standardising methods and ensuring consistent monitoring efforts worldwide. Apart from that, researchers often overlook small and cryptic individuals as less colourful, favouring larger, more visible specimens (Caldas & Robbins, 2003; Kwon et al., 2010).

It is challenging to identify insect populations. One of the reasons is that there are enormous

species of insects. Although more than a million insect species have been identified, 80% remain unnamed (Stork, 2018). Secondly, identifying insects is much more complicated than detecting vertebrates and plants, where they are hard to classify at the species level, and the identification often requires expertise equipment for genital dissections, DNA analyses and microscopic observations. Consequently, data collection on insects is inclined to collect those important in agriculture, conspicuous, and well-known, and indeed, historical data does show such bias (Wagner et al., 2021).

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Apart from the difficulties in collecting and identifying insects, it is challenging to interpret insect data. Insect population data varies between generations. The fluctuation of data makes them hard to predict, and therefore, detecting insect population change requires long-term recording of data, which is expensive and labour-intensive. In addition, most insects are specialised to the local environment, so a minor change in environmental data could affect the result (Fox et al., 2013).

CAUSES OF INSECT DECLINES

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Understanding the root causes of insect decline is vital for devising effective conservation strategies. Insects face the same threats as other organisms, such as habitat loss and degradation due to agriculture, urban development, logging, mining, and damming. Climate change, desertification, resource exploitation, environmental degradation, invasive species, chemical pollution, and atmospheric nitrification from burning fossil fuels also contribute to their decline. Unlike other organisms, such as vertebrates and plants, insects are more likely to be affected by light pollution and agricultural intensification.

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Determining which factors significantly impact insect populations is challenging due to geographical and taxonomic variance. For example, intensive agriculture is a major threat in northwestern Europe, whereas abandonment poses significant pressure in other parts of Europe (van Swaay et al., 2022). Apart from that, these factors often interact in complex and synergistic amplifying the impact on insect populations. The knock-on effects following primary threats should not be ignored, as they cause secondary consequences that impact the ecosystem. Due to space constraints, this review only covers some of these stressors and threats: habitat loss and fragmentation, pesticide use, climate change and agricultural practices.

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Habitat Loss and Fragmentation

Habitat loss and fragmentation are widely recognised as the leading causes of insect decline. The conversion of natural landscapes into agricultural and urban areas significantly reduces available habitats for insects and isolates populations, making it difficult for them to find food, mates, and shelter. Tropical forests, which host much of the world's insect biodiversity, are being cleared at alarming rates for crops, pasture, and wood fuel, particularly in Central Africa, Central America, South America, and Southeast Asia (Stork, 2018). From 2001 to 2015, an average of five million

acres of tropical forest were lost annually to industrial-scale agriculture (Stokstad, 2018). This extensive deforestation disrupts genetic diversity and increases vulnerability to environmental changes and diseases. Habitat fragmentation also isolates insect populations, making them more susceptible to local extirpation. With human populations continuing to grow, habitat destruction and fragmentation are expected to persist, emphasising the need for sustainable practices that are more nature-friendly, increase diversity, and support the ecosystem services provided by insects (Forister, Pelton & Black, 2019; Kremen & Merenlender, 2018).

Pesticide Use

The extensive use of pesticides in agriculture is another significant factor contributing to insect decline. Neonicotinoids and fipronil have drawn particular concern due to their high toxicity to many non-target insect species, including pollinators like bees and butterflies (Dudley et al., 2017; Gilburn et al., 2015; Pisa et al., 2015). Trace amounts of insecticides have been detected in wildlands, highlighting the consequences of mixed applications and sublethal effects are also troubling, such as causing stress, disorientation, or compromised immunity (Krupke et al., 2017). Neonicotinoid use has been linked to butterfly declines in the UK and California's Central Valley and is suspected in the decline of insects and insectivorous birds in the Netherlands (Gilburn et al., 2015; Forister et al., 2016). However, some insect declines predate the introduction of these insecticides and occur in areas with minimal pesticide use (Fox et al., 2013). The drift of pesticides from agricultural fields into wildlands poses significant risks, particularly in regions with extensive agricultural land use, such as Belgium, the UK, Germany, and the Netherlands, where 40–70% of land is used for agriculture.

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Climate Change

Climate change threatens insect populations by altering temperature and precipitation patterns, affecting resource availability, and shifting habitats. As we move further into the Anthropocene, climate change may prove to be one of the greatest threats to biodiversity, rivalled only by habitat destruction in tropical regions (Boggs, 2016). Insects, as poikilotherms, experience significant daily fluctuations in body temperatures, making them highly sensitive to temperature changes. While some insects may benefit from warmer temperatures, others, especially those adapted to specific climatic conditions, may suffer (Sales et al., 2018; Warren et al., 2018).

More variable weather patterns, extreme weather events, and reduced winter snow cover pose additional challenges (Ewald et al., 2015). Water stress, particularly from increasing frequency, duration, and intensity of droughts, is a significant concern, as it can cause rapid desiccation, especially during flight, and stress ecosystems that have not previously faced such moisture deficits. Changes in precipitation patterns will alter plant abundance and distribution, triggering cascading effects on dietarily specialised herbivores, which may comprise a significant portion of all metazoans (Forister et al., 2015). The warming climate also favours the spread of invasive species, which can outcompete native insects (Bertelsmeier, 2021).

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Agricultural Practices

Modern agricultural practices, including the use of monocultures and the removal of hedgerows and field margins, have drastically altered landscapes, reducing insect habitat diversity and food resources. Monocultures are particularly detrimental as they create large areas with low biodiversity unsuitable for many insect species (Tscharnkte et al., 2005). Modern farms often involve herbicide-tolerant crops, genetically modified varieties that produce insect toxins, and increased dependence on pesticides and fertilisers, all contributing to habitat degradation (Kremen, Williams & Thorp, 2002). Additionally, intensive fertiliser use can lead to nutrient runoff, further degrading habitats and affecting aquatic insect populations (Hallmann et al., 2017). Studies in northwest Germany, the UK, Finland, the Netherlands, and Sweden have identified agricultural intensification as a primary threat to insect populations, including Lepidoptera. Sustainable agricultural practices, such as agroforestry and organic farming, can help mitigate these impacts by promoting biodiversity and reducing reliance on harmful chemicals.

IMPACT OF INSECT DECLINE

The defaunation of insects means a change in biodiversity, which may largely influence ecosystem functions. A recent meta-analysis stated that the impacts on ecosystem functions from biodiversity loss are similar to those of other global changes, such as pollution and nutrient deposition. However, most efforts to quantify this relationship have primarily examined the effects of reduced producer diversity, which typically have much less impact on ecosystem functions than the loss of consumers. Apart from the implications on ecological functions, it is also worth noting that economic stability and human health might be affected.

Insects are integral to a multitude of ecological processes. Their decline leads to significant disruptions and cascading effects in ecosystem functions, particularly in pollination, nutrient cycling, and as a vital food source for higher trophic levels. Insects play a crucial role in pollination, with approximately 80% of the world's flowering plants and 75% of crop species depending on them. This reliance extends to many wild plants and crops, including fruits and vegetables, that require insect pollinators for reproduction (Ollerton, Winfree & Tarrant, 2011). The loss of pollinators thus threatens agricultural productivity and biodiversity. Nutrient cycling is another critical function supported by insects. Insects are key participants in herbivory and detritivory, breaking down organic matter and facilitating nutrient recycling. This process involves decomposing wood, leaves, dung, and carrion, returning essential nutrients to the soil (Hunter, 2001). The decline in insect populations disrupts these processes, accumulating organic matter and reducing soil fertility, which affects plant growth and ecosystem productivity (Stork et al., 2015; Bunker et al., 2005). Insects also play a pivotal role in food web dynamics. They are a primary food source for many amphibians, reptiles, mammals

and over 60% of bird species. A decrease in insect abundance can lead to trophic cascades and ecological collapses, particularly in tropical and temperate terrestrial and freshwater ecosystems (Nyffeler & Birkhofer, 2017). This defaunation accelerates the sixth great planetary extinction, significantly impacting biodiversity and ecosystem stability (Kolbert, 2014).

Insects provide economically valuable ecosystem services. The disruption of these services due to insect decline has significant economic implications. Pollination services, for example, are estimated to be worth approximately \$57 billion annually in the United States alone (Losey & Vaughan, 2006). The same report stated that the loss of pollinators would severely impact agricultural yields and food security, leading to increased costs for artificial pollination and reduced crop diversity. In addition to pollination, insects naturally control pest populations, including other invertebrates, fungi, and weeds. This reduces the need for chemical pesticides, which can be costly and environmentally harmful. The decline in insect predators and parasitoids can lead to increased pest outbreaks, higher pesticide usage, and subsequent environmental and health issues (Landis, Wratten & Gurr, 2000).

Insects also play a role in maintaining human health through various ecological functions. Many medicinal plants rely on insect pollination, and the decline in pollinators can affect the availability and diversity of these plants, impacting both traditional and modern medicine (Klein et al., 2006). Additionally, some insects act as vectors for diseases, while others help control disease vectors. Changes in insect populations can alter disease transmission dynamics, potentially increasing the prevalence of vector-borne diseases (Klein et al., 2006).

CURRENT STATE OF ACTIONS AND FUTURE DIRECTIONS

Understanding the causes and impacts of insect decline highlights the necessity for concerted efforts in conservation, policy, and research. Current efforts to mitigate insect decline include habitat restoration, pesticide regulation, and public awareness campaigns. Habitat restoration projects, such as reforestation and creating protected areas, are vital for providing environments where insect populations can thrive. For example, the UK has restored habitats to conserve butterfly species like the Duke of Burgundy (*Hamearis lucina*) and the Pearl-bordered Fritillary (*Boloria euphrosyne*; Butterfly conservation, 2022). Similarly, the DINA project in Germany focuses on restoring natural habitats in protected areas to support insect biodiversity (Köthe et al., 2023).

Regulating harmful pesticides, particularly neonicotinoids, is another critical effort. The European Union has imposed bans and restrictions on these chemicals to protect pollinators (European Commission, 2018). Raising public awareness is also crucial. Educational campaigns, citizen science projects, and community engagement initiatives play a vital role in informing and involving the public in conservation activities. Programs like the UK's Big Butterfly Count and India's National Butterfly Survey engage citizens in monitoring butterfly populations, contributing valuable data for research (Hopper & Mills, 2015; Kunte, Sondhi & Roy, 2024).

Despite these efforts, resources are often inadequate to address the broader decline in various insect species (Fox et al., 2013). Conservation actions like habitat restoration and pesticide bans have proven effective in certain contexts but often fall short due to insufficient scale and integration. For example, reforestation and pesticide regulations in Europe and the United States have helped some endangered species but struggle to address widespread pesticide residues and high carbon dioxide levels (Skendžić et al., 2021).

Future conservation efforts²⁰ must adopt integrated strategies addressing multiple threats simultaneously. Combining habitat restoration with sustainable agricultural practices and stricter pesticide regulations is essential. Integrated Pest Management (IPM) approaches are vital in reducing pesticide use through alternative pest control methods. Technological advancements, such as remote sensing and DNA metabarcoding, are crucial in monitoring insect populations and guiding conservation strategies (Stephenson, 2020). Given the significant impact of climate change on insect populations, future conservation strategies must incorporate climate adaptation measures. Creating²¹ climate-resilient habitats, facilitating species migration, and enhancing habitat connectivity are crucial for the survival of vulnerable species (Oliver & Morecroft, 2014).

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Strong policy frameworks and legislation are necessary to support these conservation efforts. Governments must implement and enforce laws protecting insect habitats, regulating pesticide use, and promoting sustainable land-use practices. International cooperation and agreements, such as the Convention on Biological Diversity, are vital to address transboundary conservation challenges and ensure a coordinated global response (Dicks et al., 2015). Engaging local communities, farmers, and other stakeholders is critical for the success of conservation initiatives. Collaborative approaches involving stakeholders in decision-making can enhance the effectiveness and sustainability of conservation actions. Building partnerships between scientists, policymakers, and communities will help create inclusive and adaptive conservation strategies²².

CONCLUSION

Insect populations are undergoing significant declines due to a combination of factors, including habitat loss, pesticide use, climate change, and agricultural practices. These declines threaten essential ecosystem services such as pollination, nutrient cycling, and pest control, with far-reaching implications for biodiversity and human welfare. Addressing this crisis requires integrated conservation strategies, including habitat restoration, stricter pesticide regulations, and sustainable agricultural practices. Additionally, advancing research methodologies and fostering international cooperation are crucial for effective conservation efforts. By understanding and mitigating the drivers of insect decline, we can work towards preserving these vital components of our ecosystems.

(3991 words)

Overall, tries to cover too much ground and requires more indepth analysis of the studies and concepts mentioned. Restructuring it to help the logic flow more smoothly and to avoid repetition would have helped reduce the word count and allowed more words for analysis. Original tables and your own original diagrams could have helped you summarise information and again save words. Also need more up to date papers.

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WHY ARE INSECTS DECLINING _ XIAOQI WU.pdf

GRADEMARK REPORT

FINAL GRADE

GENERAL COMMENTS

1 / 100

PAGE 1

PAGE 2



Comment 1

list all of them

Text Comment. this is repeated

PAGE 3



Comment 2

give details

Text Comment. why? Need more on the impacts on insects

Text Comment. Fig 1 is copied and minimally changed. Only the diagrams of the spp are missing; colours are the same too



Comment 3

how is that divided up? Decline in all species, or some?



Comment 4

how does climate change affect theses species? need to delve deeper into the papers. It's all headlines

Text Comment. This para is a list without going into details

Text Comment. Put into next section and reduce words



Comment 5

what does this mean? Explain to show your understanding



Comment 6

so species at their northern limits? Give details



Comment 7

how important is this? You need to back it up with numbers and be critical

Text Comment. Very good study to bring in, but why is this so? Is water being cleaned up? This is a key point but Lacks explanations



Comment 8

does not follow clearly



Comment 9

what does this mean? you do not expand on it below



Comment 10

How do they overcome them to be successful? Does not have any mechanistic details. reduce the papers you bring in to allow you to discuss them more deeply and analytically. These are also old papers



Comment 11

What about Van klink 2020 metanalysis of 165 studies?

Text Comment. good



Comment 12

South Africa is missing. Aquatics are not in the paper- what do they refer to? This is not clear

Title is not clear

PAGE 7



Comment 13

plants can also be very hard. This is a bit trivial



Comment 14

you've already mentioned some of them previously so this section should be earlier. Some restructuring is required so the logical flows more smoothly.



Comment 15

really? Why? This needs to be argued. Also what do you mean by agricultural intensification? Does that include the use of pesticides, fertilizers etc?



Comment 16

in what way?



Comment 17

Why? Are these the primary ones? How do you know?

PAGE 8



Comment 18

sentences need a better link. what point are you trying to make?



Comment 19

Very disjointed without a clear flow of logic

PAGE 9

Text Comment. ref needed

PAGE 10

Text Comment. how big an issue is this? Does this small section do it justice and allow you to be analytical?

PAGE 11



Comment 20

so what is the impact of this?

Text Comment. such as?



Comment 21

how???



Comment 22

30 by 30- IUCN Cop 15 should be brought in. Be up to date

Text Comment. Overall, tries to cover too much ground and requires more indepth analysis of the studies and concepts mentioned. Restructuring it to help the logic flow more smoothly and to avoid repetition would have helped reduce the word count and allowed more words for analysis. Original tables and your own original diagrams could have helped you summarise information and again save words. Also need more up to date papers.

PAGE 12

PAGE 13



Comment 23

formatting

PAGE 14

PAGE 15

PAGE 16

PAGE 17

INFORMATION C**2B**

Information Content Factual content of the text used as evidence in the background and body of dissertation. Shows grasp of the basic concepts and facts, and free of factual errors

FAIL	Fail. Not handed in or contains nothing of relevance to the subject. Content is almost wholly incorrect or shows failure to understand the task. Cannot identify topic at all
FAIL	Fail. Content demonstrates understanding of less than the expected quantity of relevant material or concepts, because of brevity, misunderstanding and/or factual errors. May be difficult to identify the topic and context.
3RD	Pass/Acceptable. Demonstrates basic understanding of more than a third of the expected amount of relevant material, but marred by significant errors and/or omissions. The topic is identifiable but shows weak grasp of context and background
2B	Good. A mostly accurate account of the subject, showing a grasp of the basic concepts and facts, but is marred by significant errors. The text gives relevant background context; a generally clear thesis statement, but with some flaws, e.g. background only weakly linked to thesis; doesn't introduce all the necessary or expected concepts.
2A	Very good. Clear grasp of the relevant concepts and facts. Gives relevant background context; clear thesis statement; relates background to that thesis
LOW 1ST	Excellent. A succinct survey of the most important relevant literature. Demonstrates excellent knowledge of context and background is all logically related to thesis
HIGH 1ST	Excellent. An excellent and succinct survey of the most important relevant literature. Any necessary mathematical, statistical or bioinformatic techniques are described logically and applied knowledgeably, Demonstrates excellent knowledge of context and background is all logically related to thesis
OUTSTANDING 1ST	Masterful/Outstanding. An outstanding and succinct survey of the most important relevant literature. Any necessary mathematical, statistical or bioinformatic techniques are described logically and applied knowledgeably.
EXCEPTIONAL 1ST	Publication quality. Dissertation shows thorough knowledge of context with ample relevant background provided linking to current state-of-the-art

CONTENT TREAT**2B**

Content Treatment Treatment of content. Use of the information in the literature sources - including Argument or Thesis, demonstrating critical/analytical/synthesis skills

FAIL	Fail. Dissertation contains nothing of relevance to the subject. Interpretation of sources is almost wholly incorrect or shows failure to understand the task
------	---

FAIL	Fail. Arguments unclear; text consists of many incorrect or irrelevant inferences and/or interpretation. Statements presented as bullet points or isolated sentences. Reveals poor comprehension of the essential concepts or of the task
3RD	Pass/Acceptable. Arguments are superficial; major omissions; major errors of interpretation of the literature
2B	Good. Shows a grasp of the basic concepts and facts, but does not go beyond that to demonstrate synthesis of ideas or sources or critical or analytical consideration of ideas. Arguments are clear but with some flaws, e.g. argument insufficiently explained; minor omissions; minor errors of comprehension; minor misinterpretation of the literature; shallow or unfocussed criticism of sources
2A	Very good. Arguments are specific and well developed; no significant errors; shows some critical, analytical, and synthetic interpretation of the literature
LOW 1ST	Excellent. Arguments are specific, well developed and defended with concrete examples; shows some critical, analytical, and synthetic skill in evaluating the sources
HIGH 1ST	Excellent. Arguments are specific, well developed and defended with concrete examples. Critical, analytical, and synthetic evaluation of the limitations of sources
OUTSTANDING 1ST	Masterful/Outstanding. Arguments are specific, well developed and defended with concrete examples; consistently critical, analytical, and synthetic in evaluating the limitations of sources
EXCEPTIONAL 1ST	Publication quality. Arguments are specific, well developed and defended with concrete examples; consistently critical, analytical, and synthetic in evaluating the limitations of sources. Conclusion concisely reinforces arguments and fully justifies the author's thesis.

STRUCTURE

2B

Text flow and organisation

FAIL	Fail. Structure is chaotic, impossible to follow. Cannot identify topic at all
FAIL	Fail. Structure is so short, or so poor that it is unclear what is being communicated; Difficult to identify the topic and context
3RD	Pass/Acceptable. Lacks a clear structure. Disorganised and inadequate paragraphing; gaps and jumps in structure make it difficult to extract meaning; large amounts of irrelevant material unrelated to the thesis; text is a literal (or figurative) list of bullet points
2B	Good. Organised paragraphs that mostly address the thesis, but with flaws, e.g. disjointed organisation of information within and between paragraphs forces reader to work out links for themselves; conclusion vague or digresses with new concepts
2A	Very good. Logically structured exposition of the subject. Paragraphs are all well organised, have topic sentences, and support the thesis; transitions within and

between paragraphs are mostly clear; conclusion sums up the arguments and author's thesis

LOW 1ST Excellent. Paragraphs are all well organised, have topic sentences, and support the thesis; transitions within and between paragraphs are clear

HIGH 1ST Excellent. Paragraphs are all well organised, consistently coherent, have topic sentences, and support the thesis; clear transitions and narrative structure throughout.

OUTSTANDING 1ST Masterful/Outstanding. Clear and succinct survey of the most important relevant primary literature. Paragraphs are all well organised, coherent, have topic sentences, and support the thesis; clear transitions and narrative structure throughout.

EXCEPTIONAL 1ST Publication quality. Paragraphs are all well organised, coherent, have topic sentences, and support the thesis; clear transitions and narrative structure throughout.

ORIGINALITY

3rd

Independent synthesis or critique, including use of figures/tables or other graphical elements to show novel or independent analysis/evaluation/insight

FAIL Fail. Contains nothing of relevance to the subject, text shows failure to understand the task (to review and analyse the literature)

FAIL Fail. Little insight into concepts presented in the literature. Graphical elements, absent, or if present then not original, or not relevant

3RD Pass/Acceptable. Some independent insight into topic and literature, but text presents material in an inconsistent, incomplete, incorrect or unscientific way that shows little original thought. Graphical elements may be original nor add meaningfully to the text

2B Good. Shows little independent insight in text or use of graphical elements. Graphical elements may not add meaningfully to the text

2A Very good. Shows some independent insight in text or use of graphical elements

LOW 1ST Excellent. States and justifies thesis in relation to literature, showing some independent insight and use of graphical elements to illustrate concepts and relationships

HIGH 1ST Excellent. Independently synthesises a structured argument, demonstrating some original insight. Excellent use of original graphical elements

OUTSTANDING 1ST Masterful/Outstanding. Independently synthesises a structured argument and is additionally original, novel, or insightful. Outstanding use of original graphical elements

EXCEPTIONAL 1ST Publication quality. Argument and presentation is original, novel, or insightful. Advanced use of original graphical elements

Quality of literature selection, including appropriate (e.g. primary vs secondary) sources

FAIL	Fail. Contains nothing of relevance to the subject - No (or almost no) information is referenced at all
FAIL	Fail. Does not show understanding of how to select, or treat literature and sources
3RD	Pass/Acceptable. Does not identify and use sufficient relevant source material. Some sources inappropriate or misused
2B	Good. Extensive reliance on non-primary sources and/or inappropriate use of sources. Some sources tangential to topic; overreliance on too few sources or on secondary sources such as reviews, webpages, or textbooks
2A	Very good. Most claims in text are substantiated and referenced. Sources appropriate and relevant, but some high-quality or important sources may be missing
LOW 1ST	Excellent. All claims in text substantiated and referenced; all sources well-chosen and of high quality
HIGH 1ST	Excellent. All claims in text are substantiated. Thoughtful selection of relevant material, identifying the most important primary literature. All sources well-chosen and of high quality.
OUTSTANDING 1ST	Masterful/Outstanding. All claims in text are substantiated appropriately. Thoughtful selection of relevant material, identifying the most important primary literature. All claims in text are substantiated appropriately
EXCEPTIONAL 1ST	Publication quality. All claims in text substantiated; all sources well chosen and of high quality; sources comprehensively cover the relevant literature

SCIENTIFIC EN

Low 1st

Scientific English Vocabulary, grammar, and accurate word usage. Clarity of words and sentences. Accessibility to intended audience

FAIL	Fail. Very poor use of English makes the dissertation almost impossible to follow
FAIL	Fail. Frequent poor use of English such that reader cannot extract meaning. Text marred by brevity, and/or errors in presentation. May be written in a non-scientific, colloquial style.
3RD	Pass/Acceptable. Consistently inaccurate use of English vocabulary, spelling, grammar, and punctuation makes it difficult to extract meaning; awkward sentences; cohesive devices showing relationships and linking ideas are used repetitively, excessively or not at all; simple sentence construction is mostly accurate but errors in more sophisticated sentences. May be written in an colloquial/unscientific style.
2B	Good. English usage may impact on clarity and meaning. Use of cohesive devices showing relationships and linking ideas is limited but mostly correct; more

sophisticated sentence structures may be attempted but are inappropriate or limited in range; may show systemic errors in e.g. subject/verb agreement, use of articles, consistent and appropriate tense/voice/mood; contains unexplained abbreviations; some jarring vocabulary or awkward phrasing. Text may be too long

2A Very good. Largely accurate use of English; any small errors do not impact on meaning; uses an appropriate range of cohesive devices to show relationships and link ideas; sophisticated sentence structures are used accurately, but only when necessary and do not impact readability

LOW 1ST **Excellent. Accurate use of English; uses a wide range of cohesive devices, sentence structures; and appropriate vocabulary to enhance readability**

HIGH 1ST Excellent. Precise, accurate, varied and sophisticated use of English; author's voice is clear

OUTSTANDING 1ST Masterful/Outstanding. Precise, accurate, varied and sophisticated use of English; author's voice is clear

EXCEPTIONAL 1ST Publication quality. Precise, accurate, varied and sophisticated use of English; author's voice is clear

FORMATTING

2B

Presentation, correct elements, literature citation and reference list style. Style should be Harvard style and must include DOI

FAIL Fail. Text, graphic elements, and references are missing, or if present are not correctly formatted; dissertation is far too long or far too short (More than twice or less than half of the expected length)

FAIL Fail. Fails to reference most sources of information. Text, graphic elements, and references formatting is incorrect and/or inconsistent; dissertation is far too long or far too short (More than twice or less than half of the expected length)

3RD Pass/Acceptable. Fails to reference at least some sources of information; some cited sources are not referenced (or vice versa). Text is poorly formatted; significantly over- or under-long; figures and tables (if present) are too basic, too detailed, too poorly constructed, or too off-topic to be meaningful or useful

2B **Good. All sources referenced but with flaws, e.g. references poorly formatted. Inconsistent formatting of headings; somewhat over- or under-long; not concise; figures and tables (if present) have poor legends, are not referred to in text, or have not been included where they would have been helpful**

2A Very good. All sources referenced using appropriate Harvard style with only minor formatting flaws. Text formatting is appropriate and mostly consistent. Keeps more-or-less to word-limit. Figures and tables (if present) are correctly formatted with good legends/titles and are referred to in text

LOW 1ST Excellent. References formatted consistently and correctly using appropriate Harvard style (consistent 'Sentence case' for titles, italic Latin names, etc.). Formatting of text and any graphical elements is consistent throughout; using

appropriate style. Graphical elements are referred to correctly in text and labelled with useful and appropriate legends/titles. Concise and keeps to word-limit

HIGH 1ST	Excellent. Sources referenced using appropriate Harvard style (consistent 'Sentence case' for titles, italic Latin names, etc.). Consistent formatting of text and any graphical elements using appropriate style. Graphical elements are correctly presented, with good legends/titles and are referred to in text. Concise and keeps to word-limit
OUTSTANDING 1ST	Masterful/Outstanding. Sources referenced using appropriate Harvard style (consistent 'Sentence case' for titles, italic Latin names, etc.). Consistent and flawless formatting of text and any graphical elements using appropriate style. Graphical elements are correctly presented, with good legends/titles and are referred to in text. Concise and keeps to word-limit
EXCEPTIONAL 1ST	Publication quality. Sources referenced using appropriate Harvard style; flawless formatting (consistent 'Sentence case' for titles, italic Latin names, etc.). Formatting and presentation of text and any graphical elements is publication-standard, and referred to in the text; elements are all the length they need to be, concise and keeps to the word-limit